

NT-C-018 | Data Structures & Algorithms | Course Knowledge Profile

Identity, purpose, prerequisites, scope and boundaries

Category	Programming & Problem Solving
Target learners	Programming learners preparing for software-development interviews and stronger implementation skills.
Prerequisites	Working knowledge of one programming language.
Approved duration	12-16 weeks / 140-180 guided hours.
Final outcome	Select, implement and explain appropriate data structures and algorithms with complexity analysis.
Assessment profile	Data/AI

Approved tools / platforms

One approved language such as Python, Java or C++; debugger, test framework, Git and coding-practice environment.

Knowledge boundaries

Include	Quality rule	Exclude / escalate
The eight approved modules, four sections, named tools, projects and mapped entry-level roles.	Require data provenance, cleaning decisions, train/test separation, suitable metrics, reproducibility and responsible interpretation.	Reject data leakage, invented metrics, unsupported causal claims and hidden manual corrections.
Current product/platform procedures only when version and date are known.	Use primary documentation and record the source version.	Do not invent current commands, prices, tax rates, policies or certification status.
Evidence-based career guidance.	Cite tests, projects, capstone, viva and verified resume evidence.	No guaranteed job, placement, salary, interview or employer claim.

LLM answer pattern: identify the course and version, answer from approved records, cite record IDs, distinguish verified facts from guidance, and state any missing evidence.

NT-C-018 | Data Structures & Algorithms | NT-C-018-S01 Deep Knowledge

Modules 1-2 | objectives, concepts, evidence and remediation

Section competency	Big-O, trade-offs and measurement; Traversal, two-pointer and sliding-window patterns
Completion evidence	Complexity worksheet; Solved problem set
Section weight	20% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 1 - Complexity

Objective: Big-O, trade-offs and measurement

Observable evidence: Complexity worksheet

Concept	Approved knowledge statement	Application behaviour	Evidence
Big-O	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when big-o is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Complexity worksheet
trade-offs	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when trade-offs is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Complexity worksheet
measurement	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when measurement is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Complexity worksheet

Module 2 - Arrays and strings

Objective: Traversal, two-pointer and sliding-window patterns

Observable evidence: Solved problem set

Concept	Approved knowledge statement	Application behaviour	Evidence
Traversal	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when traversal is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Solved problem set
two-pointer	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when two-pointer is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Solved problem set
sliding-window patterns	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when sliding-window patterns is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Solved problem set

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Complexity worksheet; Solved problem set, return to Modules 1-2, complete one guided practice and then use a new equivalent test form.

NT-C-018 | Data Structures & Algorithms | NT-C-018-S02 Deep Knowledge

Modules 3-4 | objectives, concepts, evidence and remediation

Section competency	Linked lists, stacks and queues; Recursion, divide-and-conquer and backtracking
Completion evidence	Implementations; Problem set
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 3 - Linked structures

Objective: Linked lists, stacks and queues

Observable evidence: Implementations

Concept	Approved knowledge statement	Application behaviour	Evidence
Linked lists	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when linked lists is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Implementations
stacks	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when stacks is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Implementations
queues	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when queues is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Implementations

Module 4 - Recursion

Objective: Recursion, divide-and-conquer and backtracking

Observable evidence: Problem set

Concept	Approved knowledge statement	Application behaviour	Evidence
Recursion	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when recursion is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Problem set
divide	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when divide is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Problem set
conquer	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when conquer is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Problem set
backtracking	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when backtracking is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Problem set

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Implementations; Problem set, return to Modules 3-4, complete one guided practice and then use a new equivalent test form.

NT-C-018 | Data Structures & Algorithms | NT-C-018-S03 Deep Knowledge

Modules 5-6 | objectives, concepts, evidence and remediation

Section competency	Trees, BST, traversal, priority queues; Hash maps, sets and collision concepts
Completion evidence	Tree application; Lookup solutions
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 5 - Trees and heaps

Objective: Trees, BST, traversal, priority queues

Observable evidence: Tree application

Concept	Approved knowledge statement	Application behaviour	Evidence
Trees	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when trees is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Tree application
BST	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when bst is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Tree application
traversal	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when traversal is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Tree application
priority queues	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when priority queues is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Tree application

Module 6 - Hashing

Objective: Hash maps, sets and collision concepts

Observable evidence: Lookup solutions

Concept	Approved knowledge statement	Application behaviour	Evidence
Hash maps	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when hash maps is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Lookup solutions
sets	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when sets is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Lookup solutions
collision concepts	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when collision concepts is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Lookup solutions

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Tree application; Lookup solutions, return to Modules 5-6, complete one guided practice and then use a new equivalent test form.

NT-C-018 | Data Structures & Algorithms | NT-C-018-S04 Deep Knowledge

Modules 7-8 | objectives, concepts, evidence and remediation

Section competency	Representation, BFS, DFS and shortest-path basics; Sorting, searching, greedy and dynamic programming
Completion evidence	Graph problems; Capstone problem portfolio
Section weight	30% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 7 - Graphs

Objective: Representation, BFS, DFS and shortest-path basics

Observable evidence: Graph problems

Concept	Approved knowledge statement	Application behaviour	Evidence
Representation	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when representation is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Graph problems
BFS	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when bfs is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Graph problems
DFS	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when dfs is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Graph problems
shortest-path basics	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when shortest-path basics is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Graph problems

Module 8 - Advanced patterns

Objective: Sorting, searching, greedy and dynamic programming

Observable evidence: Capstone problem portfolio

Concept	Approved knowledge statement	Application behaviour	Evidence
Sorting	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when sorting is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Capstone problem portfolio
searching	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when searching is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Capstone problem portfolio
greedy	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when greedy is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Capstone problem portfolio
dynamic programming	A defined competency within Data Structures & Algorithms. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when dynamic programming is appropriate; apply it in a Data Structures & Algorithms task; verify the result.	Capstone problem portfolio

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Graph problems; Capstone problem portfolio, return to Modules 7-8, complete one guided practice and then use a new equivalent test form.

NT-C-018 | Data Structures & Algorithms | Skill Ontology & Dependency Map

Atomic skills, learning order and evidence strength

Skill ID	Competency	Depends on	Evidence	Type
NT-C-018-SK01	Complexity	Prerequisite foundation	Complexity worksheet	Critical
NT-C-018-SK02	Arrays and strings	M01 plus earlier foundations	Solved problem set	Critical
NT-C-018-SK03	Linked structures	M02 plus earlier foundations	Implementations	Supporting
NT-C-018-SK04	Recursion	M03 plus earlier foundations	Problem set	Supporting
NT-C-018-SK05	Trees and heaps	M04 plus earlier foundations	Tree application	Supporting
NT-C-018-SK06	Hashing	M05 plus earlier foundations	Lookup solutions	Critical
NT-C-018-SK07	Graphs	M06 plus earlier foundations	Graph problems	Supporting
NT-C-018-SK08	Advanced patterns	M07 plus earlier foundations	Capstone problem portfolio	Critical

Evidence-strength hierarchy

Strength	Evidence source	Use
High	Trainer-observed practical, viva, working source/project files and reproducible output.	May support verified mastery when rubric threshold is met.
Medium	Scored mock tests, reviewed assignments and documented portfolio artifacts.	Supports mastery with corroboration.
Low	Resume claim, self-report, attendance or video completion without demonstration.	May trigger verification; cannot alone support Apply Now.

CareerTours must not treat attendance, time spent or content opened as skill mastery.

NT-C-018 | Data Structures & Algorithms | Assessment Knowledge & Sample Items

Non-live examples for LLM training and evaluation

Sample ID	Type	Question	Approved answer / rubric anchor
NT-C-018-S01-EX01	Evidence/application	What evidence best demonstrates the combined competency of Complexity and Arrays and strings?	The learner should provide Complexity worksheet; Solved problem set and explain how the work applies Big-O, trade-offs and measurement; Traversal, two-pointer and sliding-window patterns. Recall alone is insufficient.
NT-C-018-S01-EX02	Scenario/remediation	A learner reports completing Arrays and strings but cannot produce Solved problem set. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-018-S01, request Solved problem set, and allow a new equivalent test only after remediation.
NT-C-018-S02-EX01	Evidence/application	What evidence best demonstrates the combined competency of Linked structures and Recursion?	The learner should provide Implementations; Problem set and explain how the work applies Linked lists, stacks and queues; Recursion, divide-and-conquer and backtracking. Recall alone is insufficient.
NT-C-018-S02-EX02	Scenario/remediation	A learner reports completing Recursion but cannot produce Problem set. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-018-S02, request Problem set, and allow a new equivalent test only after remediation.
NT-C-018-S03-EX01	Evidence/application	What evidence best demonstrates the combined competency of Trees and heaps and Hashing?	The learner should provide Tree application; Lookup solutions and explain how the work applies Trees, BST, traversal, priority queues; Hash maps, sets and collision concepts. Recall alone is insufficient.
NT-C-018-S03-EX02	Scenario/remediation	A learner reports completing Hashing but cannot produce Lookup solutions. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-018-S03, request Lookup solutions, and allow a new equivalent test only after remediation.
NT-C-018-S04-EX01	Evidence/application	What evidence best demonstrates the combined competency of Graphs and Advanced patterns?	The learner should provide Graph problems; Capstone problem portfolio and explain how the work applies Representation, BFS, DFS and shortest-path basics; Sorting, searching, greedy and dynamic programming. Recall alone is insufficient.
NT-C-018-S04-EX02	Scenario/remediation	A learner reports completing Advanced patterns but cannot produce Capstone problem portfolio. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-018-S04, request Capstone problem portfolio, and allow a new equivalent test only after remediation.

Live bank rule: Generate at least three times the live quantity for each section using the approved Data/AI blueprint. Human-review every scored item before initial production release.

NT-C-018 | Data Structures & Algorithms | Labs, Projects & Scoring Rubrics

Practical proof of learning and ownership

Level	Approved project	Skills evidenced	Required deliverables
Mini 1	Structure library	Implement and test core structures	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Mini 2	Problem-pattern portfolio	Complexity and explanations	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Capstone	Route or scheduling solver	Graph/DP design and testing	Brief; working/source file or reproducible environment; output; validation notes; reflection.

Standard project rubric

Dimension	Weight	What earns credit
Problem framing	15%	Clear objective, user/business need, assumptions and success criteria.
Technical/design accuracy	25%	Correct methods, appropriate tools and sound decisions.
Completeness	15%	Required features, assets, data or workflow are present.
Testing and validation	15%	Important normal, edge, quality or failure cases are checked.
Evidence and reproducibility	15%	Source/working files, setup, inputs, outputs and version information are available.
Documentation	5%	Concise structure, instructions and limitations.
Viva / ownership	10%	Learner can explain decisions, demonstrate work and respond to a change request.

Require data provenance, cleaning decisions, train/test separation, suitable metrics, reproducibility and responsible interpretation.

NT-C-018 | Data Structures & Algorithms | Career Roles, Evidence & Gap Actions

Explainable top-three occupation matching

Rank	Occupation	Default weighted skills	Minimum evidence	Gap action
1	Software Engineer Trainee	DSA 34% [critical]; testing 33% [critical]; explanation 33%	Reviewed problem portfolio	If a critical skill is below 60 or reviewed problem portfolio is missing, return Improve First or Needs Evidence with the linked module/project.
2	Problem-Solving Trainee	Patterns 50% [critical]; complexity 50% [critical]	Timed assessments	If a critical skill is below 60 or timed assessments is missing, return Improve First or Needs Evidence with the linked module/project.
3	Backend Developer Trainee	Algorithms 50% [critical]; implementation 50% [critical]	Capstone solver	If a critical skill is below 60 or capstone solver is missing, return Improve First or Needs Evidence with the linked module/project.

Course readiness	45% section mocks + 20% mini projects + 25% capstone + 10% viva.
Role readiness	45% role-relevant mastery + 25% projects/capstone + 20% aligned mocks + 10% verified resume evidence.
Evidence confidence	Show separately; Apply Now requires at least 70.
Resume extraction	Use only relevant skills, projects, education, experience and portfolio evidence; minimise personal data before external model processing.

A student may be Apply Now for one mapped occupation and Improve First for another. Return independent evidence and gaps for each role.

NT-C-018 | Data Structures & Algorithms | FAQs, Misconceptions & LLM Response Pack

Course-specific retrieval and response patterns

Approved FAQ	Answer
Who is Data Structures & Algorithms for?	Programming learners preparing for software-development interviews and stronger implementation skills.
What should I know before starting?	Working knowledge of one programming language.
How long is the approved learning journey?	12-16 weeks / 140-180 guided hours.
What will I be able to demonstrate?	Select, implement and explain appropriate data structures and algorithms with complexity analysis.
Does completion guarantee a job?	No. Completion and role readiness are evidence-based guidance; employment, placement, salary and employer selection are not guaranteed.
What if I fail a section?	CareerTours identifies the failed skill tags and module, assigns targeted practice, and permits an equivalent retest after remediation.

Misconception	Approved correction
Attendance proves mastery	False - require tests and observable practical/project evidence.
One total score proves every role fit	False - calculate each occupation using its weighted skills and evidence.
The LLM can add missing details	False - retrieve approved records or state that evidence is insufficient.
Course completion guarantees employment	False - never claim guaranteed job, placement, salary or interview.
Data Structures & Algorithms knowledge never changes	False - version-sensitive tools, laws, taxes, security and platform procedures require review.

Retrieval metadata

```
course_code=NT-C-018; corpus_version=3.0;
record_types=course_profile,module,section,skill,assessment_sample,project,occupation_mapping,faq,misconception;
approval_status=approved; effective_date=2026-08-12
```